Right Triangles & Trigonometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 97°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (31^\circ + 52^\circ) = 97^\circ$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 103°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (32^\circ + 45^\circ) = 103^\circ$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Level 2 · MediumIn a right triangle ABC where angle $C = 90^\circ$, $\cos(A) = \frac{5}{13}$. What is the value of $\tan(A)$?

In right triangle ABC, $\cos(A) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{5}{13}$. Using Pythagoras: opposite $= \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. $\tan(A) = \frac{\text{opposite}}{\text{adjacent}} = \frac{12}{5} = 2.4$.

4 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

In a right triangle, acute angle θ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

Step 1: In a right triangle with opposite side 5 and hypotenuse 13, the adjacent side is: $\text{adj} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$. Step 2: Compute trigonometric ratios:$\cos\theta = \frac{\text{adj}}{\text{hyp}} = \mathbf{\frac{12}{13}}$. $\tan\theta = \frac{\text{opp}}{\text{adj}} = \mathbf{\frac{5}{12}}$.$\sec\theta = \frac{1}{\cos\theta} = \mathbf{\frac{13}{12}}$.

5 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

In right triangle ABC with $C = 90^\circ$, $a = 5$, and $c = 13$. What is the value of $\sin A + \cos A$?

By the Pythagorean theorem, the adjacent leg is $b = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = 12$. Then $\sin A = \frac{a}{c} = \frac{5}{13}$ and $\cos A = \frac{b}{c} = \frac{12}{13}$. Adding them together: $\sin A + \cos A = \frac{5+12}{13} = \mathbf{\frac{17}{13}}$.

6 **[Foundational]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Level 3 · AdvancedThe equation of a circle in the xy-plane is $x^2 - 8x + y^2 + 6y = 24$. What are the coordinates of the center and the radius of the circle?

Complete the square for both x and y: $(x^2 - 8x + 16) + (y^2 + 6y + 9) = 24 + 16 + 9$ $(x-4)^2 + (y+3)^2 = 49$. In standard form $(x-h)^2 + (y-k)^2 = r^2$, the center is $(h, k) = (4, -3)$ and the radius is $r = \sqrt{49} = 7$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 4 · SynthesisThe equation of a circle in the xy-plane is $x^2 + y^2 - 10x + 6y - 15 = 0$. What is the length of the diameter of the circle?

Complete the square for both x and y: $(x^2 - 10x + 25) + (y^2 + 6y + 9) = 15 + 25 + 9 = 49$.
$(x - 5)^2 + (y + 3)^2 = 7^2$. The radius is $r = 7$, so the diameter is $2r = 14$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsA cone-shaped storage tank has a base radius of 4 feet and a height of 9 feet. What is the volume of the tank in cubic feet, rounded to the nearest tenth?

Using the cone volume formula from the reference sheet: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4)^2(9) = \frac{1}{3}\pi(16)(9) = 48\pi \approx 150.8$ cubic feet.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

Step 1: Invert the fraction to remove the negative exponent: $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4} = \left(\frac{81z^{12}}{16x^8y^{-4}}\right)^{3/4} = \left(\frac{81y^4z^{12}}{16x^8}\right)^{3/4}$.
Step 2: Take the 4th root of each factor: $\left(\frac{81y^4z^{12}}{16x^8}\right)^{1/4} = \frac{81^{1/4}(y^4)^{1/4}(z^{12})^{1/4}}{16^{1/4}(x^8)^{1/4}} = \frac{3yz^3}{2x^2}$. Step 3: Cube the result.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.

Step 1: Convert to degrees: $\theta = \frac{7\pi}{6} \times \frac{180^\circ}{\pi} = 7 \times 30^\circ = \mathbf{210^\circ}$. Step 2: Terminal coordinates in Quadrant III: The reference angle is $210^\circ - 180^\circ = 30^\circ$. In Quadrant III, both coordinates are negative: $x = \cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}$, $y = \sin(210^\circ) = -\sin(30^\circ) = -\frac{1}{2}$. The terminal point is $\left(\mathbf{-\frac{\sqrt{3}}{2}, -\frac{1}{2}}\right)$.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedIn the system of equations below, k is a constant. For what value of k does the system have no solution? $\{kx - 6y = 15; 4x - 3y = 9$

For a system of two linear equations in standard form $Ax + By = C$ to have no solution, the lines must be parallel (equal ratios of x- and y-coefficients, but unequal ratio of constants): $\frac{k}{4} = \frac{-6}{-3} eq \frac{15}{9}$ $\frac{k}{4} = 2 \Rightarrow k = 8$.
Check constant ratio: $\frac{15}{9} = \frac{5}{3} eq 2$, confirming the lines never intersect. Thus, $k = 8$.

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12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

Recall the basic trigonometric ratio from SOH CAH TOA: $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{3}{5} = \mathbf{0.6}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a+bi$.

Step 1: Expand the numerator: $(3-2i)^2 = 3^2 - 2(3)(2i) + (2i)^2 = 9 - 12i + 4(-1) = 5 - 12i$. Step 2: Multiply by conjugate $1-i$: $\frac{5-12i}{1+i} = \frac{(5-12i)(1-i)}{(1+i)(1-i)} = \frac{5-5i-12i+12i^2}{1-i^2} = \frac{5-17i-12}{1-(-1)} = \frac{-7-17i}{2}$. In standard form: $\mathbf{-\frac{7}{2} - \frac{17}{2}i}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

(a) Use the Cosine Rule: $XZ^2 = 7^2 + 10^2 - 2(7)(10)\cos(65^\circ) = 49 + 100 - 140(0.4226) = 149 - 59.166 = 89.834$

$XZ = \sqrt{89.834} \approx 9.4779\ldots \approx 9.48$ cm (3 s. f.) (b) Area of triangle:

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}(7)(10)\sin(65^\circ) = 35(0.9063) \approx 31.7$ cm^2 (3 s. f.)

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.01 Foundations & Reasoning**

A right triangle has $\tan\theta = \frac{3}{4}$. Find $\sin\theta$.

Step 1: Determine the triangle sides: Since $\tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{3}{4}$, set opposite $= 3$ and adjacent $= 4$. Step 2: Find hypotenuse: $c = \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$ Step 3: Compute $\sin\theta$: $\sin\theta = \frac{\text{opp}}{\text{hyp}} = \mathbf{\frac{3}{5} = 0.6}$

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 10.01 Coordinate Geometry**

If $\cos(2x+10^\circ) = \sin(3x-15^\circ)$, find the value of x and evaluate $\tan(x+16^\circ)$.

Step 1: Apply cofunction identity $\cos A = \sin B \Leftrightarrow A + B = 90^\circ$:

$(2x+10^\circ) + (3x-15^\circ) = 90^\circ \Rightarrow 5x - 5^\circ = 90^\circ \Rightarrow 5x = 95^\circ \Rightarrow \mathbf{x = 19^\circ}$. Step 2: Evaluate tangent:

$\tan(19^\circ + 16^\circ) = \tan(35^\circ) \approx \mathbf{0.7002}$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability

Which of the following expressions is equivalent to $\frac{1-\sin^2\theta}{\cos\theta}$ for all values of θ where the expression is defined?

Using the fundamental Pythagorean identity $\sin^2\theta+\cos^2\theta=1$, we have $1-\sin^2\theta=\cos^2\theta$. Substituting this into the numerator: $\frac{1-\sin^2\theta}{\cos\theta}=\frac{\cos^2\theta}{\cos\theta}=\boldsymbol{\cos\theta}$

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

If $\cos\theta=-\frac{4}{5}$ and θ lies in Quadrant II, what is the value of $\tan\theta$?

In Quadrant II, $x=-4$ and hypotenuse $r=5$. Using $x^2+y^2=r^2$:
$(-4)^2+y^2=5^2\Rightarrow 16+y^2=25\Rightarrow y=3$ (positive in QII) Therefore: $\tan\theta=\frac{y}{x}=\frac{3}{-4}=\boldsymbol{-\frac{3}{4}}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

If $A=(2,\ -1;3,4)$ and $B=(5;\ -2)$, what is the matrix product AB?

The product of a 2×2 matrix and a 2×1 matrix is a 2×1 column vector. Compute row by column: First row: $2(5)+(-1)(-2)=10+2=12$. Second row: $3(5)+4(-2)=15-8=7$. Thus $AB=(\mathbf{12;7})$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Find the hypotenuse of a right triangle with legs 7 and 24.

Step 1: Apply the Pythagorean theorem: $c=\sqrt{a^2+b^2}=\sqrt{7^2+24^2}$ Step 2: Compute squares and sum: $c=\sqrt{49+576}=\sqrt{625}=\mathbf{25}$ (This is the classic $7-24-25$ Pythagorean triple). The hypotenuse is 25.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying angles of triangles, $m\angle A=54^\circ$ and $m\angle B=45^\circ$. What is $m\angle C$?

a. 81° ✓

b. 91°

c. 73°

d. 126°

Using the triangle angle-sum theorem, $m\angle C=180^\circ-(54^\circ+45^\circ)=81^\circ$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 105° ✓

b. 115°

c. 97°

d. 150°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (30^\circ + 45^\circ) = 105^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95° ✓

b. 105°

c. 87°

d. 147°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (33^\circ + 52^\circ) = 95^\circ$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right triangle ABC, the measure of $\angle C$ is 90°. If $\sin A = \frac{x+1}{13}$ and $\cos B = \frac{3x-7}{13}$, what is the value of $\tan A$?

a. $\frac{5}{12}$ ✓

b. $\frac{12}{5}$

c. $\frac{5}{13}$

d. $\frac{12}{13}$

Step 1: Apply the complementary cofunction identity. In right triangle ABC with right angle at C, the acute angles satisfy $A + B = 90^\circ$. Therefore, $\cos B = \cos(90^\circ - A) = \sin A$. Step 2: Solve for x. Equating $\sin A$ and $\cos B$: $\frac{x+1}{13} = \frac{3x-7}{13}$ $x + 1 = 3x - 7 \Rightarrow 2x = 8 \Rightarrow x = 4$ Step 3: Find the trigonometric ratios for angle A. Substitute $x = 4$ into the sine expression: $\sin A = \frac{4+1}{13} = \frac{5}{13}$ Since.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Right triangle legs 3 and 4: hypotenuse:

a. 3

b. 6

c. 7

d. **5** ✓

Applying Triangles principles, $c = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$. Therefore, the verified correct option is 5.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A right triangle has legs 12 and 16 and hypotenuse 20. What is its area?

a. 48

b. 106

c. **96** ✓

d. 192

Area $= \frac{1}{2} \times 12 \times 16 = 96$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A right circular cone has a base radius of 5 and a height of 12. What is the lateral area of the cone?

a. 60π

b. 65π ✓

c. 90π

d. 130π

To find the lateral area of a right circular cone, use the formula: $\text{Lateral Area} = \pi r l$ where r is the base radius and l is the slant height. First, find the slant height l using the Pythagorean relationship between radius r, height h, and slant height l: $l = \sqrt{r^2 + h^2} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ Now, substitute $r = 5$ and $l = 13$ into the lateral area formula: $\text{Lateral Area} = \pi(5)(13) = 65\pi$ Common misconceptions: - Using the vertical.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right circular cylinder has a radius of 5 cm and a height of 10 cm. What is the total surface area of the cylinder?

a. 150π cm^2

b. 250π cm^2

c. **350π cm^2** ✓

d. 500π cm^2

The formula for the total surface area of a right circular cylinder is $TSA = 2\pi r^2 + 2\pi rh$, where r is the radius and h is the height. Given $r = 5$ cm and $h = 10$ cm, we substitute these values into the formula: $TSA = 2\pi(5)^2 + 2\pi(5)(10)$ $TSA = 2\pi(25) + 2\pi(50)$ $TSA = 50\pi + 100\pi$ $TSA = 150\pi$ cm^2.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two equilateral triangles have side lengths in the ratio $4:9$. What is the ratio of their altitudes?

a. $2:3$

b. $16:81$

c. $8:27$

d. **$4:9$ ✓**

All equilateral triangles have interior angles of 60°, so all equilateral triangles are similar by AA. All corresponding linear dimensions (sides, altitudes, medians, perimeters) scale in the exact same linear ratio k. Therefore, the ratio of their altitudes is $4:9$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right ΔABC with right angle at C, altitude $CD \perp AB$. If $AD = 3$ and $DB = 12$, what is the length of leg AC?

a. **$3\sqrt{5} \approx 6.71$ ✓**

b. 6

c. 15

d. $6\sqrt{5} \approx 13.42$

Total hypotenuse is $AB = AD + DB = 3 + 12 = 15$. By the Geometric Mean Leg Theorem, each leg is the geometric mean of the hypotenuse and the adjacent segment: $AC^2 = AB \cdot AD$. Substituting: $AC^2 = 15 \times 3 = 45 \Rightarrow AC = \sqrt{45} = 3\sqrt{5} \approx 6.71$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?

a. 100π

b. $\frac{400\pi}{3}$ ✓

c. $\frac{500\pi}{3}$

d. 200π

Sphere radius is $r = 5$. Cylinder volume $V_c = \pi r^2 h = \pi(25)(12) = 300\pi$. Sphere volume $V_s = \frac{4}{3}\pi(5^3) = \frac{500\pi}{3}$. Space outside sphere: $300\pi - \frac{500\pi}{3} = \frac{900\pi - 500\pi}{3} = \frac{400\pi}{3}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In right ΔABC with right angle at C, altitude CD is drawn perpendicular to hypotenuse AB. If $AD = 4$ and $DB = 9$, what is the length of altitude CD?

a. 6.5

b. **6** ✓

c. 5.2

d. 7.2

By the Geometric Mean Altitude Theorem, the altitude to the hypotenuse is the geometric mean of the two segments into which it divides the hypotenuse: $CD^2 = AD \cdot DB$. Substituting: $CD^2 = 4 \times 9 = 36 \Rightarrow CD = \sqrt{36} = 6$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔJKL with right angle at vertex K, an altitude from K meets hypotenuse JL at point M. Given that segment $JM = 5$ and $ML = 15$, determine the length of leg JK.

a. $5\sqrt{3}$

b. $10\sqrt{3}$

c. **10** ✓

d. $5\sqrt{5}$

By the Geometric Mean Leg Theorem (projection theorem): $JK^2 = JM \cdot JL$. The full hypotenuse is $JL = JM + ML = 5 + 15 = 20$. Therefore, $JK^2 = 5 \times 20 = 100 \Rightarrow JK = \sqrt{100} = 10$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right ΔABC with $\angle C = 90^\circ$, $AC = 6$, $BC = 8$, and hypotenuse $AB = 10$. The angle bisector of $\angle A$ intersects side BC at point D. What is the exact length of segment CD?

a. 4

b. 2.5

c. **3** ✓

d. 3.5

By the Angle Bisector Theorem applied to angle A: $\frac{CD}{DB} = \frac{AC}{AB} = \frac{6}{10} = \frac{3}{5}$. Since $CD + DB = BC = 8$, $CD = 8 \times \frac{3}{3+5} = 8 \times \frac{3}{8} = 3$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In right ΔABC, altitude CD to hypotenuse AB has length 12. If segment $AD = 9$, what is the length of hypotenuse AB?

a. 16

b. 20

c. 21

d. **25** ✓

By the Geometric Mean Altitude Theorem: $CD^2 = AD \cdot DB \Rightarrow 12^2 = 9 \cdot DB \Rightarrow 144 = 9 \cdot DB \Rightarrow DB = 16$. The total hypotenuse is $AB = AD + DB = 9 + 16 = 25$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar triangles have a linear scale factor of $3 : 7$. If the perimeter of the smaller triangle is 24, what is the perimeter of the larger triangle?

a. 48

b. 72

c. 130.67

d. **56** ✓

Perimeters of similar figures scale linearly with the scale factor k: $\frac{P_2}{P_1} = \frac{7}{3} \Rightarrow P_2 = 24 \times \frac{7}{3} = 56$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a right triangle, the legs are 5 and 12. What is the hypotenuse?

a. **13** ✓

b. 15

c. 17

d. $\sqrt{119}$

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $\sqrt{25 + 144} = \sqrt{169} = 13$
Therefore, the verified correct option is 13.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

In a right triangle, the legs have lengths 9 and 12. What is the area of the triangle?

a. 27

b. **54 ✓**

c. 72

d. 108

Area $= \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times 9 \times 12 = 54$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

All equilateral triangles are equiangular; this triangle is equilateral, so it is equiangular. This is an example of:

a. Inductive reasoning

b. A logical fallacy

c. Circular reasoning

d. **Deductive reasoning ✓**

Deductive reasoning applies a general rule or definition to a specific case to reach a certain conclusion.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

All right angles measure 90°; $\angle A$ is a right angle, so $\angle A = 90^\circ$. This is an example of:

a. A logical fallacy

b. Inductive reasoning

c. Circular reasoning

d. **Deductive reasoning ✓**

Deductive reasoning applies a general rule or definition to a specific case to reach a certain conclusion.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A student folds several isosceles triangles in half and finds the fold is always a line of symmetry, then concludes this holds for every isosceles triangle. This is an example of:

a. Deductive reasoning

b. **Inductive reasoning ✓**

c. Circular reasoning

d. A logical fallacy

Inductive reasoning draws a general conclusion from a limited number of specific observations or patterns.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles

What is the contrapositive of the statement: "If Two triangles are congruent, then their corresponding sides are equal."?

a. If Their corresponding sides are equal, then two triangles are congruent.

b. If Two triangles are not congruent, then their corresponding sides are equal.

c. **If Their corresponding sides are not equal, then two triangles are not congruent.** ✓

d. If Two triangles are not congruent, then their corresponding sides are not equal.

The contrapositive swaps and negates both parts — it is always logically equivalent to the original.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume

What is the contrapositive of the statement: "If Two lines are perpendicular, then they form a right angle."?

a. If Two lines are not perpendicular, then they do not form a right angle.

b. **If They do not form a right angle, then two lines are not perpendicular.** ✓

c. If Two lines are not perpendicular, then they form a right angle.

d. If They form a right angle, then two lines are perpendicular.

The contrapositive swaps and negates both parts — it is always logically equivalent to the original.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

Angles 40° and 60°: third angle:

a. 100

b. 90

c. 70

d. **80** ✓

Applying Triangles principles, $180 - 40 - 60 = 80^\circ$. Therefore, the verified correct option is 80.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

Isosceles triangle with two sides 7 and base 10: perimeter:

a. 21

b. 17

c. **24** ✓

d. 26

Applying Triangles principles, $P = 2(7) + 10 = 24$. Therefore, the verified correct option is 24.

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